Q1. What is the difference between single-quoted and double-quoted strings in Python?

Ans:

There is no difference in functionality. Both are used to create strings.

- Single quotes: 'Hello'
- Double quotes: "Hello"

Example:

a = 'Hello'

b = "Hello"

print(a)

print(b)

Both produce the same output: Hello.

When to use: Use double quotes when your string contains an apostrophe, such as "I'm happy".

Q2. What is the difference between mutable and immutable objects?

Ans:

- Mutable: Objects whose values can be changed after creation.
- Immutable: Objects whose values cannot be changed after creation.

Examples:

- Mutable: List, Dictionary, Set
- Immutable: String, Tuple, Integer, Float

```
# Mutable example
data = [10, 20, 30]
data[0] = 50
print(data)  # [50, 20, 30]

# Immutable example
name = "Python"
# name[0] = "J"  # This gives an error
```

Q3. What is the difference between a list and a tuple in Python?

Ans:

List	**Tuple**
Mutable (changeable)	Immutable (unchangeable)
Uses square brackets []	Uses parentheses ()
Elements can be added or removed	Elements cannot be added or removed
Example: [10, 20, 30]	Example: (10, 20, 30)

Use case:

- List: When data needs to be changed.
- Tuple: When data should remain fixed.

Q4. What is the difference between a set and a list in terms of functionality and use cases?

Ans:

Set	**List**
Stores unique elements	Allows duplicate elements
Unordered collection	Maintains insertion order
Uses curly brackets {}	Uses square brackets []
Does not support indexing	Supports indexing
Useful for removing duplicates	Useful for storing ordered data

Example:

```
my_list = [1, 2, 2, 3]

my_set = {1, 2, 2, 3}


print(my_list)  # [1, 2, 2, 3]

print(my_set)   # {1, 2, 3}
```

5. How does a dictionary differ from a list in terms of data storage and retrieval?

Ans:

Dictionary	**List**
Stores data in key-value pairs	Stores individual elements
Uses curly brackets {}	Uses square brackets []
Values are accessed using keys	Values are accessed using indexes
Keys must be unique	Duplicate values are allowed

Example:

```
# Dictionary
student = {"name": "Apurva", "age": 22}
print(student["name"])


# List
data = ["Apurva", 22]
print(data[0])
```

Output:

Apurva

Apurva